TABLE I

MAIN MODEL PARAMETERS IN THE DEFAULT CASE STUDY.

Symbol	Description	Value
Cfix	Fixed infrastructure cost	153600 $
Ccons_sl	Installation cost per slow EVSE	540.07 $/EVSE
Ccons_fa	Installation cost per fast EVSE	25000 $/EVSE
Ctrans	Transportation cost scalar	0.0435 $/km
Cunmet	Unmet charging-demand penalty	3 $/kWh
CLS_cri/noncri	Critical/non-critical load-shedding penalty	50/10 $/kWh
gamma	Discount rate	0.01 -
theta	Payback period	20 year
pi_f	Disaster-regime weight	0.3 -
P_EV_sl/fa	Rated charging power of slow/fast EVSE	7/50 kW
Nmax_sl/fa	Maximum slow/fast EVSE per EVCS	25/10 -
K	Default outage budget	2 lines
—A—, —B—	Default common evaluator support	10, 10 scenarios

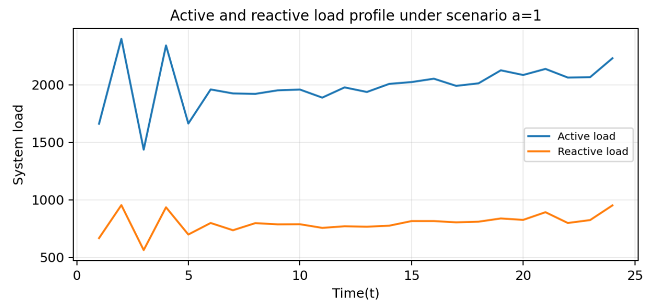

Fig. 1. Active and reactive load profile under representative normal scenario $a = 1$.

I. CASE STUDIES

A. Experimental Setup

The case study is performed on the IEEE 33-node radial distribution feeder. The feeder contains critical and non-critical load buses, and EV charging stations can be placed at candidate buses along the main feeder and laterals. Each opened station may install slow and fast EVSEs, with the station-level upper bounds given in Table I. The normal-operation stage models 24 hours of EV charging service and distribution-system operation. The disaster stage models the 10:00–14:00 emergency window, in which available EV discharging capability is used to support loads after line outages.

Table I lists the parameters used in the default study. The default uncertainty support contains ten normal scenarios and ten disaster operating scenarios. The outage ambiguity set uses budget $K = 2$, so the adversarial replay can select two failed lines when evaluating disaster resilience. The reported objective components are always computed after the first-stage siting and sizing decisions are fixed and replayed under this common evaluator.

B. Scenario Generation and Uncertainty Support

The normal scenarios $a \in A$ contain the time-varying load and regional EV charging-demand profiles. The disaster scenarios $b \in B$ contain the post-disaster load and available V2G discharging profiles. The line-outage uncertainty is not treated as another sampled scenario. Instead, the outage pattern is chosen by the DRO separation problem subject to the budget K. Thus B describes disaster operating conditions, while K controls the adversarial line-outage support.

Figs. 1–3 show representative profiles from the same audited 10×10 data source used in the default optimization and replay. The slow-charging demand has a broad daily profile, whereas fast charging is more concentrated in short peaks. The disaster V2G profiles are defined on the four-period emergency window, which creates a different operating tradeoff from ordinary charging service.

C. Benchmark Definitions

Four cases are compared. Case 1 is the proposed integrated DRO planner, which jointly optimizes annualized construction cost, normal-operation recourse, and disaster-stage DRO recourse. Case 2 is the normal-operation-only benchmark, which removes the disaster term and therefore represents the daily-service endpoint. Case 3 is the disaster-resilience-only benchmark, which removes the normal-stage EV service objective and constraints during training; the resulting plan is then replayed under the normal evaluator to show the daily-service consequence of a disaster-only design. Case 4 is a fair deterministic mean-value benchmark. It replaces the multi-scenario normal and disaster supports by their mean profiles, but solves the same integrated planning model with the same outage budget, $K_{\text{train}} = 2$.

All four first-stage plans are frozen and evaluated under the same $|A| = 10$, $|B| = 10$, $K_{\text{eval}} = 2$ MILP worst-distribution evaluator. Hence the comparisons below are not comparisons of four different training objectives. They are common-scale ex-post comparisons of the resulting plans. The training objective for the accepted default uses normalized top-level multipliers $(m_{\text{cons}}, m_{\text{nor}}, m_{\text{dis}}) = (0.0152, 1, 1.4)$, while all reported costs in the tables are raw physical/economic components without these multipliers.

D. Default Component Comparison

Fig. 4 shows the three partial-framework first-stage plans, and Table II reports the corresponding component costs. The table follows the same decomposition as the reference case-study structure: F^{cons} is the construction component, F^{trans} is EV user transportation cost, F^{unmet} is unmet charging-demand penalty, F^{sub} is substation purchase cost, $\Psi^{\text{nor}} = F^{\text{trans}} + F^{\text{unmet}} + F^{\text{sub}}$ is the normal operation component, and Φ^{dis} is the worst-distribution disaster component.

Case 2 demonstrates the normal-service endpoint. It opens only five stations and achieves nearly the same normal component as the proposed plan: Ψ^{nor} is 62,280.70 in Case 2 and 62,324.28 in Case 1. This small normal-operation difference is expected because both plans preserve low transportation cost and zero unmet charging demand. The difference appears when the plans are replayed under disaster uncertainty. Case 2 covers only 1/11 critical buses directly, and its Φ^{dis} is 16,774.96. The integrated plan increases construction cost and opens broader feeder coverage, but reduces Φ^{dis} to 5,223.70, a 68.86% reduction.

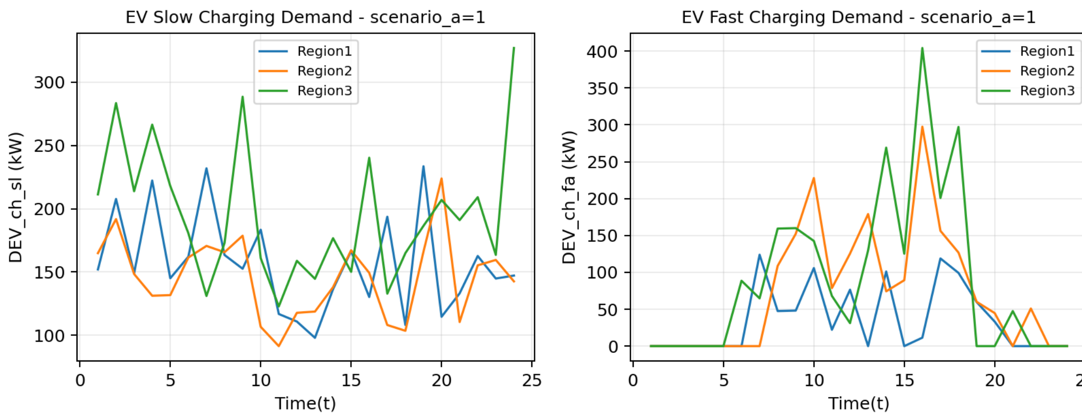

Fig. 2. Slow and fast EV charging demand profiles for each region under representative normal scenario $a = 1$.

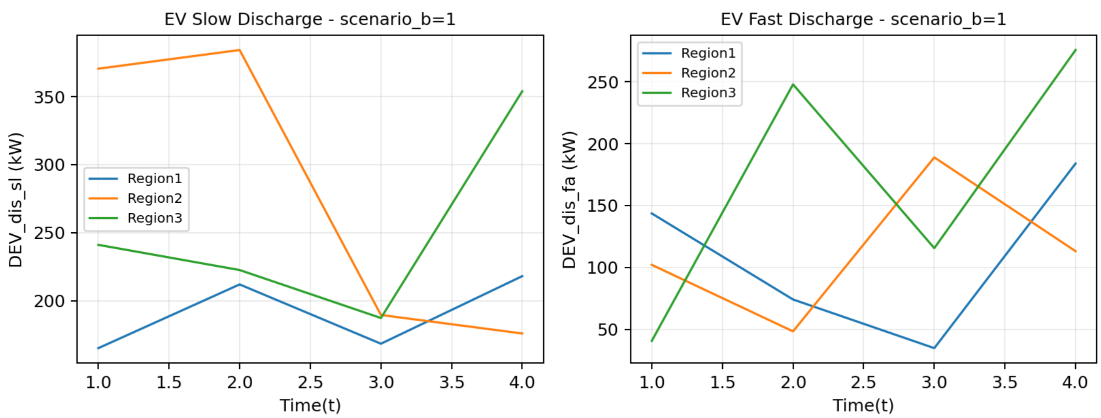

Fig. 3. Slow and fast EV discharging availability profiles for each region under representative disaster scenario $b = 1$.

Case 3 gives the opposite endpoint. It opens thirteen stations, directly covers 9/11 critical buses, and has the lowest disaster component, $\Phi^{\mathrm{dis}} = 4,358.42$. This confirms that the disaster-only benchmark is indeed the most resilience-oriented design. However, that design is not a good daily-service plan. Under the same normal replay, F^{unmet} rises to 66,167.24 and accounts for 50.91% of Ψ^{nor}. The integrated plan therefore sits between the two partial frameworks: it keeps zero unmet demand like the normal-only endpoint, while recovering most of the disaster benefit of the disaster-only endpoint.

The topology in Fig. 4 explains this tradeoff. Case 1 keeps daily-service anchors at buses 3, 6, 30, and 32, but adds buses 8, 11, 14, 17, 18, 21, 26, and 29 relative to Case 2 while dropping bus 19. These additional stations spread capacity along the main radial backbone and the two laterals. In physical terms, the integrated model does not simply add chargers at the cheapest daily-demand locations. It shifts siting toward critical-bus neighborhoods and the far upper lateral, increasing direct critical-bus coverage from 1/11 to 7/11. This is the mechanism behind the component table: the plan pays for wider first-stage coverage, not for a large change in daily EV transportation service.

E. Deterministic Worst-Distribution Comparison

Case 4 isolates the value of multi-scenario DRO support from the value of the outage budget itself. The deterministic plan is trained on mean normal and mean disaster profiles, but it still uses $K_{\mathrm{train}} = 2$. Table III then freezes the proposed and deterministic plans and replays both under the same $|A| = 10$, $|B| = 10$, $K_{\mathrm{eval}} = 2$ evaluator.

The deterministic solution is a meaningful comparator because it has the same outage-budget semantics as the proposed model. Its weakness is that the mean profiles do not represent the multi-scenario tails used in the common replay. It opens ten stations and installs 101 slow and 12 fast EVSEs, whereas the proposed DRO plan opens twelve stations and installs 172 slow and 15 fast EVSEs. The additional capacity in Case 1 is not concentrated at a single bus: it expands coverage around buses 17, 21, and 26 and strengthens the upper lateral through buses 29, 30, and 32. Under the same worst-distribution replay, this broader topology reduces Φ^{dis} from 6,545.72 to 5,223.70, which is a 20.20% reduction. The evaluated total cost also falls from 196,871.27 for the deterministic plan to 173,264.00 for the proposed plan, because the deterministic first-stage saving

TABLE II
OBJECTIVE COMPONENTS UNDER THE COMMON $|A| = 10$, $|B| = 10$, $K = 2$ MILP WORST-DISTRIBUTION EVALUATOR.

Obj. ($)	Case 1	Case 2	Case 3	Case 4
F^{cons}	128,069.89	66,721.59	132,216.04	104,765.26
F^{trans}	21,514.49	21,470.91	23,014.97	21,837.60
F^{unmet}	0.00	0.00	66,167.24	66,167.24
F^{sub}	40,809.78	40,809.78	40,769.87	40,769.87
Ψ^{nor}	62,324.28	62,280.70	129,952.07	128,774.70
Φ^{dis}	5,223.70	16,774.96	4,358.42	6,545.72
J_{eval}	173,264.00	115,350.56	224,490.02	196,871.27
Sites	12	5	13	10
Slow/Fast EVSE	172/15	113/15	165/12	101/12
Critical buses covered	7/11	1/11	9/11	5/11

TABLE III
PROPOSED AND FAIR DETERMINISTIC MEAN-VALUE PLANS UNDER THE COMMON WORST-DISTRIBUTION EVALUATOR.

Metric	Proposed	Det. $K_{\mathrm{train}} = 2$
F^{cons}	128,069.89	104,765.26
Ψ^{nor}	62,324.28	128,774.70
Φ^{dis}	5,223.70	6,545.72
$\pi_f \Phi^{\mathrm{dis}}$	1,567.11	1,963.72
J_{eval}	173,264.00	196,871.27
Sites	12	10
Slow/Fast EVSE	172/15	101/12
Critical buses covered	7/11	5/11
Reduction in Φ^{dis}	–	20.20%

TABLE IV
CERTIFICATION RECORD FOR THE DEFAULT CASE.

Case	Certificate	Iter.	Final viol.
Case 1	ϵ-certified	13	94.96
Case 2	exact	0	0.00
Case 3	ϵ-certified	34	95.07
Case 4	ϵ-certified	17	49.99

is outweighed by poorer common replay performance.

F. Certification and Reproducibility

Table IV reports the certification status for the four default runs. The normal-only benchmark is solved directly to optimality. The three DRO/Benders runs reach the ϵ-certified stopping rule under the same $|A| = 10$, $|B| = 10$, $K = 2$ support. For traceability, the component rows, plans, replay logs, and figure sources are retained with the experiment pack; rerunning the recorded instance reproduces all entries in Table II, including station counts, charger counts, critical-bus coverage values, and open-bus sets.

The resulting default case supports the main claim of this section: compared with normal-only planning, the integrated DRO model purchases broader critical coverage with almost no normal-service degradation; compared with disaster-only planning, it avoids the large unmet-demand penalty; and compared with a fair mean-value deterministic planner using the same outage budget, it achieves lower worst-distribution disaster cost under the common replay.

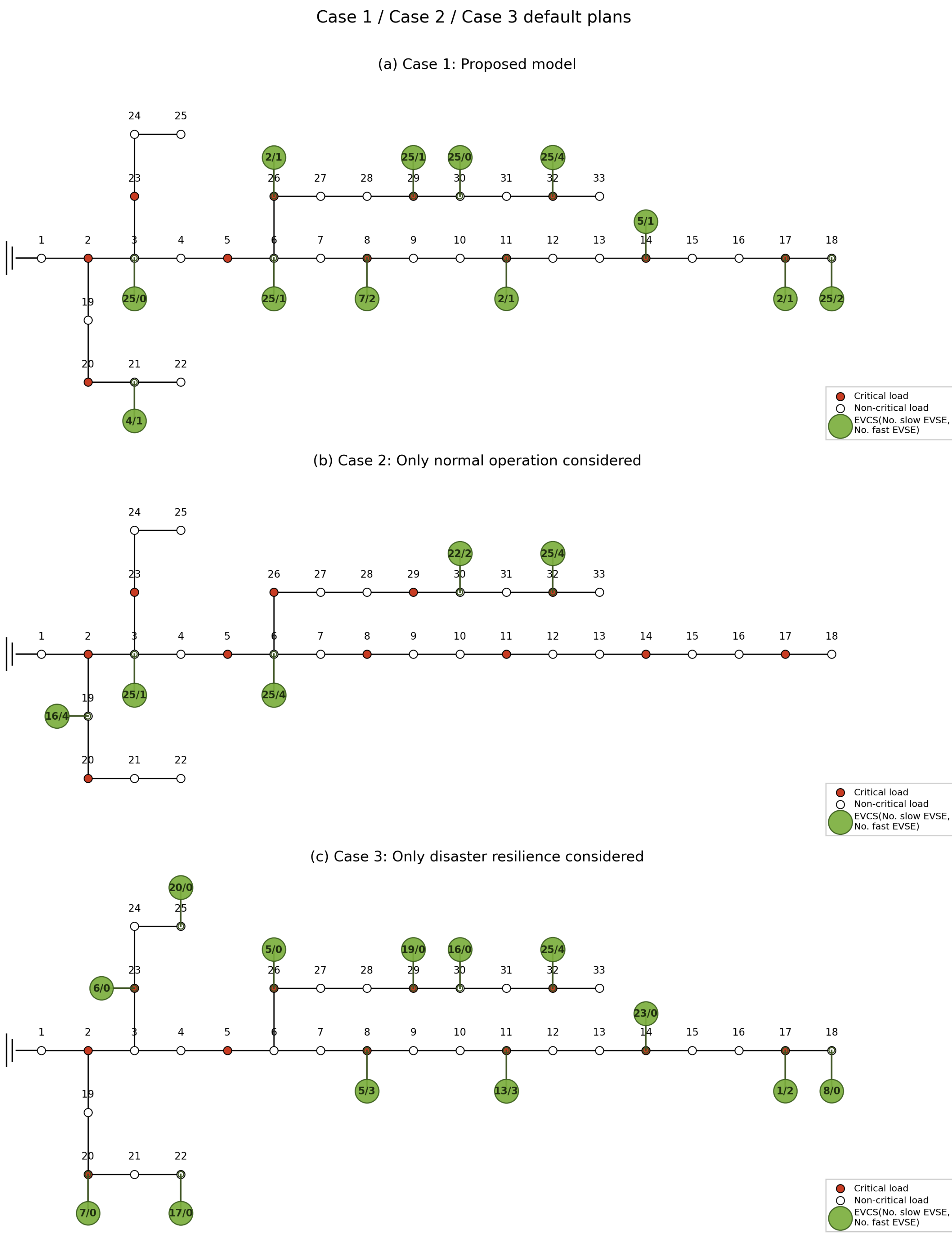

Fig. 4. Planning maps for the proposed integrated, normal-only, and disaster-only cases.

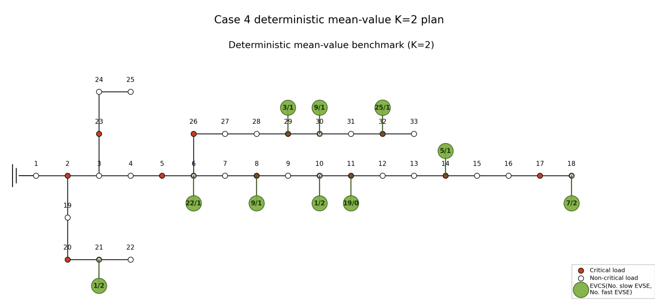

Fig. 5. Fair deterministic mean-value planning map using mean normal and disaster supports with $K_{\mathrm{train}} = 2$.